

Chapter 9 – Types of Cyber Attacks & Social Engineering

Introduction to Cyber Attacks

A **Cyber Attack** is any attempt by a hacker or attacker to:

- Steal data 
- Damage systems 
- Gain unauthorized access 
- Spy on users 
- Disrupt services 

Attackers use different methods such as:

- Malware
- Social Engineering
- Password attacks
- Network attacks
- Web attacks

Cyber attacks can target:

- Computers 
 - Mobile phones 
 - Servers 
 - Networks 
 - Websites 
 - Cloud systems 
-

1. Malware Attack

What is Malware?

Malware means **Malicious Software**.

It is software intentionally created to:

- Damage systems
- Steal information
- Spy on users
- Encrypt files
- Destroy data
- Gain unauthorized access

Simple Definition

Any harmful software used by hackers is called malware.

Types of Malware Attacks

1. Virus Attack

Definition

A **Virus** is malware that attaches itself to files or programs and spreads when the infected file is opened.

How Virus Works

1. Virus attaches to a legitimate file
2. User opens the file
3. Virus becomes active
4. It starts infecting other files

Features

- Requires user interaction
- Can corrupt files
- Can slow down the system
- May delete data

Example

A fake game installer .exe file contains a virus.

When the user installs the game:

- Virus activates
- Files become infected

Prevention

- Use antivirus
- Avoid unknown software
- Don't open suspicious attachments
- Keep OS updated

Daily Life Example

Imagine a rotten apple  placed among fresh apples.

Slowly all apples become rotten.

That is how a virus spreads.

2. Worm Attack

◆ Definition

A **Worm** is self-replicating malware that spreads automatically without user action.

How Worm Works

- Spreads through:
 - Networks
 - USB drives
 - Emails
 - Shared folders
 - It automatically copies itself to other systems.
-

Important Point

Unlike viruses:

- Worms do NOT require users to open files.
-

Characteristics

- Spreads very fast
 - Consumes bandwidth
 - Slows networks
 - Hard to remove
-

Example

ILOVEYOU Worm (2000)

Spread through email attachments and infected millions of systems.

Prevention

- Update antivirus
 - Close unused ports
 - Disable autorun
 - Use firewall
-

Daily Life Example

Like a contagious disease 🤒 spreading automatically from one person to another.

3. Ransomware Attack

◆ Definition

Ransomware locks or encrypts files and demands money to unlock them.

How It Works

1. System gets infected
 2. Files become encrypted 
 3. Hacker demands ransom money 
 4. Payment usually requested in Bitcoin
-

Why Bitcoin?

Because:

- Anonymous
 - Hard to trace
-

Famous Examples

- WannaCry
 - Petya
-

Effects

- Files cannot be opened
 - Businesses stop working
 - Huge financial loss
-

Prevention

- Take backups
 - Avoid suspicious attachments
 - Use ransomware protection
 - Keep systems updated
-

Daily Life Example

Like someone locking your house and asking money for the key .

4. Bot Attack

◆ Definition

A **Bot** is an automated malicious program.

Many infected bots together form a:

Botnet

What Bots Do

Bots can:

- Send spam emails
 - Launch DDoS attacks
 - Spread malware
 - Mine cryptocurrency
-

How It Works

1. Device infected
 2. Connects to hacker's command server (C&C)
 3. Hacker controls thousands of infected devices
-

Example

Mirai Botnet

Used IoT devices for massive DDoS attacks.

Daily Life Example

Like an army of robots  following one master's command.

5. Trojan Attack

Definition

A Trojan is malware disguised as legitimate software.

How Trojan Works

- Looks safe
 - User installs it
 - Malware activates secretly
-

Trojan Can:

- Install ransomware
 - Spy on users
 - Steal files
 - Give attacker remote access
-

Important Point

Trojan does NOT self-replicate.

Example

Fake:

- Game installer
 - Cracked software
 - System cleaner
-

Daily Life Example

Like a gift box 📺 containing a bomb inside.



6. Spyware

◆ Definition

Spyware secretly monitors user activities.



Spyware Can:

- Record mic 🎤
 - Access webcam 📷
 - Monitor browsing
 - Steal passwords
 - Track location
-



Example

Pegasus Spyware

Created by NSO Group.

Can:

- Record calls
 - Read messages
 - Access phone data
-



Prevention

- Avoid suspicious apps
 - Limit permissions
 - Use anti-spyware tools
-

Daily Life Example

Like someone secretly following and observing you .

7. Keylogger

◆ Definition

A Keylogger records every key pressed on the keyboard.

Captures

- Passwords
 - Banking details
 - Messages
 - Emails
-

How It Works

Runs silently in background and sends logs to attacker.

Prevention

- Use MFA
 - Avoid unknown extensions
 - Use virtual keyboard for banking
-

Daily Life Example

Like someone writing down every word you type .

8. Wiper Malware

◆ Definition

A Wiper permanently deletes files and destroys data.

Purpose

- Destroy systems
 - Remove evidence
 - Cause damage
-

Example

Shamoon Attack

Destroyed data on thousands of systems.

Daily Life Example

Like burning all documents in an office  .

9. Cryptojacking

◆ Definition

Using someone else's computer secretly to mine cryptocurrency.

How It Works

- Malicious script runs in background
 - CPU/GPU usage becomes high
 - Hacker earns crypto
-

Symptoms

- System becomes slow
 - Laptop heats up
 - Fan runs continuously
-

Prevention

- Use antivirus
 - Block malicious scripts
 - Monitor CPU usage
-

10. Phishing

◆ Definition

Fake emails/messages designed to steal information.

Example

“Your Instagram password expired. Click here.”

Fake page steals login credentials.

Goal

- Steal passwords
 - Banking details
 - OTPs
-

Prevention

- Check URLs carefully
- Don't share OTP

- Use MFA



Daily Life Example

Like a fake bank employee asking for ATM PIN.



11. Spoofing



Definition

Pretending to be someone else.



Types

- Email Spoofing
- SMS Spoofing
- Caller ID Spoofing



Purpose

To gain trust and steal information.



12. Password Attack



Definition

Trying to crack passwords using tools or guessing techniques.



Methods



Brute Force

Trying all possible passwords.

Dictionary Attack

Using common password lists.

Social Engineering

Guessing using personal information.

Prevention

- Strong passwords
 - Use symbols & numbers
 - Enable 2FA
-

13. DDoS Attack

Definition

Distributed Denial of Service attack floods a server with huge traffic.

How It Works

Thousands of infected devices send fake requests.

Server becomes overloaded.

Result

Website becomes unavailable.

Example

Gaming and banking website attacks.

Prevention

- Firewalls
 - Cloudflare
 - Rate limiting
-

Daily Life Example

Like thousands of fake customers entering a shop so real customers cannot enter.

14. Code Injection Attack

◆ Definition

Injecting malicious code into applications.

SQL Injection Example

```
SELECT * FROM users WHERE username=" OR '1'='1' --;
```

This can bypass login authentication.

Attacker Can:

- Access database
 - Delete data
 - Modify records
-

Prevention

- Input validation
- Prepared statements
- Sanitization

15. Cross-Site Scripting (XSS)

◆ Definition

Injecting malicious JavaScript into webpages.

Example

```
<script>alert('Hacked')</script>
```

Effects

- Steal cookies
 - Redirect users
 - Execute actions
-

Prevention

- Sanitize inputs
 - Use CSP
 - Escape HTML
-

16. Logic Bomb

◆ Definition

Malicious code that activates when a condition is met.

Examples

- Script activates after resignation
- Trigger on specific date

Prevention

- Code reviews
 - Monitoring
 - Access control
-

Social Engineering

◆ **What is Social Engineering?**

Manipulating humans instead of hacking systems.

Hackers exploit:

- Trust
 - Fear
 - Urgency
 - Emotions
-

Types of Social Engineering Attacks

Phishing

Fake emails/websites to steal credentials.

Spear Phishing

Targeted phishing using personal details.

Business Email Compromise (BEC)

Fake executive emails requesting money transfer.

Vishing

Voice phishing through calls.

Smishing

SMS phishing.

Pretexting

Creating fake stories to obtain information.

Baiting

Using infected USBs or fake offers.

Quid Pro Quo

Offering help in exchange for credentials.

Tailgating / Piggybacking

Following authorized users into restricted areas.

Water-Holing

Compromising websites frequently visited by targets.

Deepfake Attacks

Using fake AI-generated voice/video.

Social Engineering Tools

GUIDED BY ANISH NAIR SIR (B7)

Tool	Purpose
SET	Phishing & fake login pages
Evilginx2	Advanced phishing
Gophish	Email phishing campaigns
Bettercap	MITM attacks
Fluxion	Fake Wi-Fi attacks
MSFvenom	Malicious payloads



Common Targets

- HR employees
- Finance department
- Customer support
- Home users



Protection Against Social Engineering

Protection	Purpose
Awareness Training	Identify scams
MFA	Extra security
Email Filtering	Block malicious emails
Caller Verification	Verify identity
Antivirus	Detect malware
HTTPS Websites	Secure communication

Real-World Case Study – Target Data Breach (2013)

Attackers compromised an HVAC vendor and entered Target Corporation systems.

Result:

- 40 million credit cards stolen

Lesson:

One weak vendor can compromise an entire company.

Shoulder Surfing Attack

◆ Definition

Watching someone's screen or keyboard to steal information.

Example

Watching ATM PIN while standing behind someone.

Prevention

- Use privacy screen
 - Cover keyboard
 - Avoid entering passwords publicly
-

Man-in-the-Middle (MITM) Attack

◆ Definition

Attacker secretly sits between two communicating devices.

How It Works

Victim ↔ Attacker ↔ Router/Website

Attacker can:

- Read data
 - Modify traffic
 - Steal credentials
-

Example

Public Wi-Fi attacks at cafes.

Prevention

- Use HTTPS
 - Use VPN
 - Avoid public Wi-Fi
-

ARP Poisoning Attack

Definition

A MITM technique where attacker sends fake ARP messages.

Goal

Victim thinks:

- Hacker's MAC = Router's MAC

So all traffic goes through attacker.

Attacker Can

- Capture passwords
- Monitor traffic
- Modify packets

Prevention

- Static ARP entries
- Use HTTPS
- Network monitoring

Final Important Notes

Malware Types

- Virus
- Worm
- Trojan
- Spyware
- Ransomware
- Keylogger
- Wiper
- Cryptojacking

Social Engineering Attacks

- Phishing
 - Vishing
 - Smishing
 - BEC
 - Tailgating
 - Shoulder Surfing
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